Retrieval-Augmented Generation:
Baseline Dense Retrieval vs. Cross-Encoder Re-Ranking

Assignment Report

Name: Shantanu Mande Roll No.:2026701007

Code repository: https://github.com/shantanumande/rag-reranking-warhammer

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Document: Warhammer 40,000 Core Rules (294 chunks).

Bi-encoder: BAAI/bge-small-en-v1.5.

Cross-encoder: BAAI/bge-reranker-base.

LLM: nvidia/nemotron-3.5-super-vl (OpenRouter), identical prompt for both pipelines.

Retrieval: top 10 candidates; top 3 passed to the LLM.

1. ARCHITECTURE COMPARISON

A bi-encoder encodes the query and each chunk separately into fixed vectors (384 dimensions for bge-small). Chunk vectors are computed once in advance (15.9 s for 294 chunks); at query time only the query is encoded (~34 ms) and compared with every chunk by cosine similarity. This is fast, but each chunk is summarized without knowing the question, so the model judges topical similarity rather than whether a chunk actually answers the query.

A cross-encoder takes the query and a chunk as one input sequence, so self-attention links every query word to every chunk word. It outputs a single relevance score for that pair, reflecting whether the chunk answers the question, not just whether it shares vocabulary. The cost is that nothing can be precomputed: every (query, chunk) pair needs a full model pass at query time.

Table 1. Bi-encoder and cross-encoder compared.

Property	Bi-encoder	Cross-encoder
Model used	BAAI/bge-small-en-v1.5	BAAI/bge-reranker-base
Input	Query and chunk, encoded separately	Query and chunk together, as one sequence
Relevance score	Cosine similarity of two vectors	Single score from joint attention
Chunk representations precomputed	Yes, once	No; every pair at query time

Property	Bi-encoder	Cross-encoder
Strength	Fast; scales to large collections	Accurate; models how query and chunk relate

2. QUALITATIVE AND RANK-SHIFT ANALYSIS

A failure query is one where the chunk containing the answer was ranked 4–10 by the bi-encoder (so it was not passed to the LLM in Pipeline 1) and moved into the top 3 by the cross-encoder (so it was passed to the LLM in Pipeline 2).

Table 2. Rank changes of the critical chunk for each failure query.

Failure query	Critical chunk	Pipeline 1 rank	Pipeline 2 rank
Lance (keyword trap)	wh-129 WEAPON ABILITIES › LANCE	#7	#1
Pistol (exception to a general rule)	wh-125 WEAPON ABILITIES › PISTOL	#6	#2
Precision (keyword trap)	wh-133 WEAPON ABILITIES › PRECISION (summary)	#4	#2

2.1 Lance

Query: "Do any weapons get a bonus to their wound roll if the unit charged this turn?"

Table 3. Lance query: context passed to the LLM and the resulting answers.

	Pipeline 1 (baseline)	Pipeline 2 (re-ranked)
Top 3 chunks	wh-139 Heavy (summary), wh-151 Charge Bonus, wh-120 Assault (summary)	wh-129 Lance, wh-207 Tank Shock, wh-212 Heroic Intervention
Answer	"The provided rules do not cover this."	Lance weapons add 1 to the Wound roll if the bearer made a Charge move that turn [wh-129].
Outcome	No answer given, although the rule exists	Correct, quoted from the rule

The bi-encoder favoured chunks sharing surface words with the query although none grants a Wound roll bonus. For example "charge" (Charge Bonus), "weapon" and "bonus" (Heavy, Assault). Lance, the only correct rule, sat at #7. With no relevant chunk in its context, the LLM refused to answer. This is a retrieval failure.. The cross-encoder scored Lance 0.94 and moved it to #1.

2.2 Pistol

Query: "My unit is in Engagement Range of an enemy. Is there any weapon it can still shoot with?"

Table 4. Pistol query: context passed to the LLM and the resulting answers.

	Pipeline 1 (baseline)	**Pipeline 2 (re-ranked)**
Top 3 chunks	wh-092 Big Guns Never Tire (Designer's Note), wh-088 Locked in Combat, wh-091 Big Guns Never Tire	wh-092 Big Guns Never Tire (Designer's Note), wh-125 Pistol, wh-088 Locked in Combat
Overall verdict	"Generally, no"	"Yes"
Only Pistols may be fired	Omitted	Stated
Only one enemy unit within Engagement Range may be targeted	Omitted	Stated
Outcome	Essential clauses omitted	Rule stated with its restrictions

The query's wording ("Engagement Range", "shoot") matches the general prohibition (wh-088) almost exactly, while the Pistol exception sits in a separate section with different wording, so it ranked #6. Answer 1 therefore framed the rule as "generally no", mentioned Pistols only via a Designer's Note, and omitted both restrictions: only Pistols may be fired, and only at an enemy unit within Engagement Range. Re-ranking moved Pistol to #2 and Answer 2 stated the rule correctly, though it covered the Monster/Vehicle exception more briefly once wh-091 dropped out.

2.3 Precision

Query: "My attack wounded an enemy unit led by a Character. Can I allocate that wound to the Character?" The bi-encoder ranked wh-101 (ALLOCATE ATTACK) first; the cross-encoder moved wh-133 (PRECISION) from #4 to #2.

The bi-encoder ranked the general allocation rule (wh-101) and the Leader rule (wh-197) highest, leaving the Precision rule at #4. Answer 1 therefore stated that a wound on an Attached unit can never be allocated to its Character, omitting the Precision exception entirely and presenting the restriction as absolute. Re-ranking promoted Precision to #2, and Answer 2 gave the complete rule: allocation to the Character is normally forbidden, but a Precision attack can be allocated to a visible Character, as can remaining attacks once the last Bodyguard model is destroyed. As with Pistol, the baseline omitted an essential exception rather than inventing a rule.

	Pipeline 1 (baseline)	**Pipeline 2 (re-ranked)**
Top 3 chunks	wh-101 Allocate Attack, wh-197 Leader, wh-102 Allocate Attack (summary)	wh-197 Leader, wh-133 Precision (summary), wh-101 Allocate Attack
Attached unit: allocate the wound to the Character?	"No", stated as absolute	"No", unless an exception applies
Precision exception (attacker may allocate to a visible Character)	Omitted	Stated, quoted from wh-133

	Pipeline 1 (baseline)	**Pipeline 2 (re-ranked)**
Exception once the last Bodyguard model is destroyed	Omitted	Stated, from wh-197
Who allocates in the normal case	Unclear	The defending player
Outcome	Essential exception omitted	Complete rule with both exceptions

Across 31 evaluation queries, the correct chunk was in the top 3 for 20 with the bi-encoder and 23 after re-ranking. Five queries never reached the top 10, so re-ranking could not help: it improves precision, but recall is set by the first stage. Re-ranking also occasionally hurt (Armour Penetration fell from #1 to #3).

3. LATENCY PROFILING

Measured on CPU over the three failure queries, five runs each.

Table 5. Runtime of the baseline retrieval and the re-ranking pipeline.

Stage	**Median**	**Minimum**	**Maximum**
Pipeline 1: bi-encoder retrieval (top 10)	34.2 ms	25.3 ms	47.9 ms
Cross-encoder re-ranking alone (10 pairs)	3,820.7 ms	3,191.2 ms	4,800.7 ms
Pipeline 2: retrieval + re-ranking	3,867.1 ms	3,227.4 ms	4,834.3 ms
LLM answer generation (either pipeline)	18.6 s	3.4 s	109.5 s

One-off indexing cost (bi-encoder, all 294 chunks): 15.9 s.

Re-ranking makes retrieval about 113× slower (3.87 s vs. 34 ms), because the cross-encoder runs ten full passes of a larger model over query and chunk together, instead of encoding the query once. Even so, it is about a fifth of the median LLM time (18.6 s), whose wide range (3.4–109.5 s) reflects free-tier queueing rather than the model itself.

4. SCALING QUESTION

Measured: running the cross-encoder over all 294 chunks for one query took **62.0 s**, against 3.8 s for the top 10 candidates.

A cross-encoder cannot precompute anything, so every query costs one full model pass per chunk. Scoring all 294 chunks took 62 s (≈ 0.21 s per chunk); for one million chunks that is about 2.4 days per query. A bi-encoder embeds the collection once offline, and each query is then a single encoding plus a fast vector search. The two-stage design gets the best of both: the bi-encoder cheaply narrows the collection to 10 candidates, and the accurate cross-encoder ranks only those. The trade-off is that a chunk missed in the first stage can never be recovered.